

Aim

To write a Java program to display each character of a string using the charAt() method.

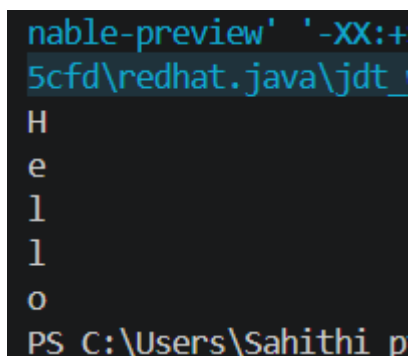
Algorithm

1. Start the program.
2. Declare and initialize a string s with "Hello".
3. Find the length of the string using s.length().
4. Use a for loop from 0 to s.length() - 1.
5. Access each character using s.charAt(i).
6. Display each character on a separate line.
7. Stop the program.

Code:

```
package Assignment;  
  
public class CharaAt {  
  
    public static void main(String[] args) {  
  
        String s = "Hello";  
  
        for(int i=0;i<s.length();i++){  
  
            System.out.println(s.charAt(i));  
  
        }  
  
    }  
  
}
```

Output:



```
nale-preview' '-XX:+  
5cfd\redhat.java\jdt_  
H  
e  
l  
l  
o  
PS C:\Users\Sahithi p
```

Concatenation:

Aim

To write a Java program to concatenate two strings using the + operator.

Algorithm

1. Start the program.
2. Declare and initialize the string firstName with "Hello".
3. Declare and initialize the string lastName with "Java".
4. Concatenate both strings with a space between them.
5. Store the result in the result variable.
6. Display the concatenated string.
7. Stop the program.

Code:

```
package Assignment;

public class Concatenation {

    public static void main(String[] args) {

        String firstName = "Hello";

        String lastName = "Java";

        String result = firstName + " " + lastName;

        System.out.println(result);

    }

}
```

Output:

```
Assignments.Concatenation'
Hello Java
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>
```

Aim:

To write a Java program to check whether a specified sequence of characters is present in a string using the contains() method.

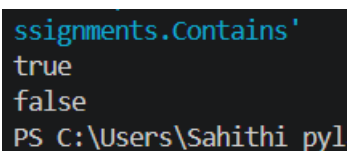
Algorithm

1. Start the program.
2. Declare and initialize a string s with "Hello World".
3. Use the contains() method to check whether "Hello" is present in the string.
4. Display the result.
5. Use the contains() method to check whether "java" is present in the string.
6. Display the result.
7. Stop the program.

Code:

```
package Assignments;  
  
public class Contains {  
    public static void main(String[] args) {  
        String s = "Hello World";  
        System.out.println(s.contains("Hello"));  
        System.out.println(s.contains("java"));  
    }  
}
```

Output:



```
Assignments.Contains'  
true  
false  
PS C:\Users\Sahithi pyl
```

Aim

To write a Java program to compare two strings using the equals() method.

Algorithm

1. Start the program.

2. Declare and initialize str1 with "Java".
3. Declare and initialize str2 with "Java".
4. Use the equals() method to compare str1 and str2.
5. If both strings are equal, display **"Strings are equal"**.
6. Otherwise, display **"Strings are not equal"**.
7. Stop the program.

Code:

```
package Assignments;

class StringEquals {

    public static void main(String[] args) {

        String str1 = "Java";

        String str2 = "Java";

        if (str1.equals(str2)) {

            System.out.println("Strings are equal");

        } else {

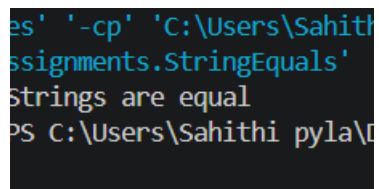
            System.out.println("Strings are not equal");

        }

    }

}
```

Output:



```
es' '-cp' 'C:\Users\Sahithi\Assignments.StringEquals'
Strings are equal
PS C:\Users\Sahithi\pyla\l
```

Aim

To write a Java program to find the position of a specified substring in a string using the indexOf() method.

Algorithm

1. Start the program.
2. Declare and initialize the string str with "Hello Java".
3. Use the indexOf() method to find the position of "Java".
4. Store or directly display the returned position.
5. Print the position of "Java".
6. Stop the program

Code:

```
package Assignments;

public class IndexOf {

    public static void main(String[] args) {

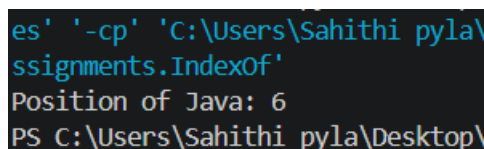
        String str = "Hello java";

        System.out.println("Position of Java: " + str.indexOf("Java"));

    }

}
```

Output:



```
es' '-cp' 'C:\Users\Sahithi\pyla\
ssignments.IndexOf'
Position of Java: 6
PS C:\Users\Sahithi\pyla\Desktop\
```

Aim

To write a Java program to check whether a string is empty using the isEmpty() method.

Algorithm

1. Start the program.
2. Declare and initialize an empty string str.
3. Use the isEmpty() method to check whether the string is empty.
4. If isEmpty() returns true, display **"String is empty"**.
5. Otherwise, display **"String is not empty"**.

6. Stop the program

Code:

```
package Assignment;

class StringIsEmpty {

    public static void main(String[] args) {

        String str = "";

        if (str.isEmpty()) {

            System.out.println("String is empty");

        } else {

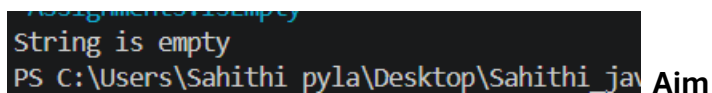
            System.out.println("String is not empty");

        }

    }

}
```

Output:



```
String is empty
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_jav Aim
```

To write a Java program to convert all the characters of a string into lowercase using the `toLowerCase()` method.

Algorithm

1. Start the program.
2. Declare and initialize the string `str` with "HELLO JAVA".
3. Use the `toLowerCase()` method to convert the string into lowercase.
4. Display the converted string.
5. Stop the program

Code:

```
package Assignments;
```

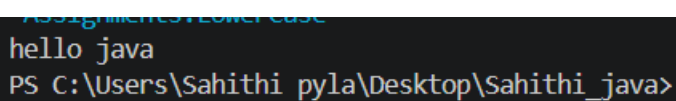
```

class LowerCase {
    public static void main(String[] args) {
        String str = "HELLO JAVA";

        System.out.println(str.toLowerCase());
    }
}

```

Output:



```

Assignment:Lower Case
hello java
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>

```

Aim

To write a Java program to check whether a string contains only numbers using the matches() method and regular expression.

Algorithm

1. Start the program.
2. Declare and initialize the string str with "12345".
3. Use the matches() method with the regular expression [0-9]+.
4. If the string contains only digits, display **"String contains only numbers"**.
5. Otherwise, display **"String contains other characters"**.
6. Stop the program.

Code:

```

package Assignment;

class StringMatches {
    public static void main(String[] args) {
        String str = "12345";

        if (str.matches("[0-9]+")) {
            System.out.println("String contains only numbers");
        }
    }
}

```

```

    } else {
        System.out.println("String contains other characters");
    }
}
}
}

```

Output:

```

'Assignments.Matches'
String contains only numbers
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Aim:

To write a Java program to replace a specified substring with another substring using the `replace()` method.

Algorithm

1. Start the program.
2. Declare and initialize the string `str` with "I like Java".
3. Use the `replace()` method to replace "Java" with "Python".
4. Store the replaced string in the result variable.
5. Display the result.
6. Stop the program

```
package Assignment;
```

```

class StringReplace {
    public static void main(String[] args) {
        String str = "I like Java";

        String result = str.replace("Java", "Python");

        System.out.println(result)
    }
}

```

```
}  
}
```

Aim:

To write a Java program to split a string into individual words using the split() method.

Algorithm

1. Start the program.
2. Declare and initialize the string str with "Java is easy".
3. Use the split(" ") method to split the string based on spaces.
4. Store the words in a string array words.
5. Use a for-each loop to access each word.
6. Display each word on a separate line.
7. Stop the program.

Code:

```
package Assignment;
```

```
class StringSplit {  
    public static void main(String[] args) {  
        String str = "Java is easy";  
  
        String[] words = str.split(" ");  
  
        for (String word : words) {  
            System.out.println(word);  
        }  
    }  
}
```

Output:

```
nmments.Split'  
Java  
is  
easy  
PS C:\Users\Sahithi_pyla\Desktop\Sahit
```

Aim

To write a Java program to convert all the characters of a string into uppercase using the toUpperCase() method.

Algorithm

1. Start the program.
2. Declare and initialize the string str with "hello java".
3. Use the toUpperCase() method to convert the string into uppercase.
4. Display the converted string.
5. Stop the program.

Code:

```
package Assignment;
```

```
public class UpperCase {
```

```
    public static void main(String[] args) {
```

```
        String str = "hello java";
```

```
        System.out.println(str.toUpperCase());
```

```
    }
```

```
}
```

Output:

```
nmments.UpperCase'  
HELLO JAVA  
PS C:\Users\Sahithi_pyla\Deskt
```

Aim

To write a Java program to find the length of a string using the length() method.

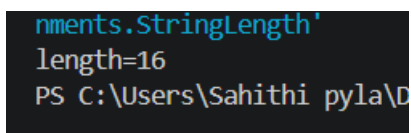
Algorithm

1. Start the program.
2. Declare and initialize the string s with "java programming".
3. Use the length() method to find the number of characters in the string.
4. Display the length of the string.
5. Stop the program.

Code:

```
package Assignments;  
  
public class StringLength {  
  
    public static void main(String[] args) {  
  
        String s="java programming";  
  
        System.out.println("length=" + s.length());  
  
    }  
  
}
```

Output:



```
Assignments.StringLength'  
length=16  
PS C:\Users\Sahithi\pyla\D
```

Aim

To write a Java program to check whether a string starts with a specified prefix and ends with a specified suffix using startsWith() and endsWith() methods.

Algorithm

1. Start the program.
2. Declare and initialize the string str with "Hello Java".
3. Use the startsWith() method to check whether the string starts with "Hello".
4. Display the result.
5. Use the endsWith() method to check whether the string ends with "Java".

6. Display the result.

7. Stop the program

Code:

```
package Assignment;
```

```
public class StringStartEnd {
```

```
    public static void main(String[] args) {
```

```
        String str = "Hello Java";
```

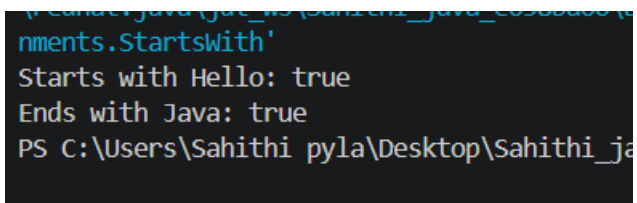
```
        System.out.println("Starts with Hello: " + str.startsWith("Hello"));
```

```
        System.out.println("Ends with Java: " + str.endsWith("Java"));
```

```
    }
```

```
}
```

Output:

A screenshot of a terminal window showing the output of a Java program. The text is as follows:
C:\Users\Sahithi> java -cp . Assignment\StringStartEnd
Starts with Hello: true
Ends with Java: true
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>